Technical Report: Personalized Netflix Recommendation System

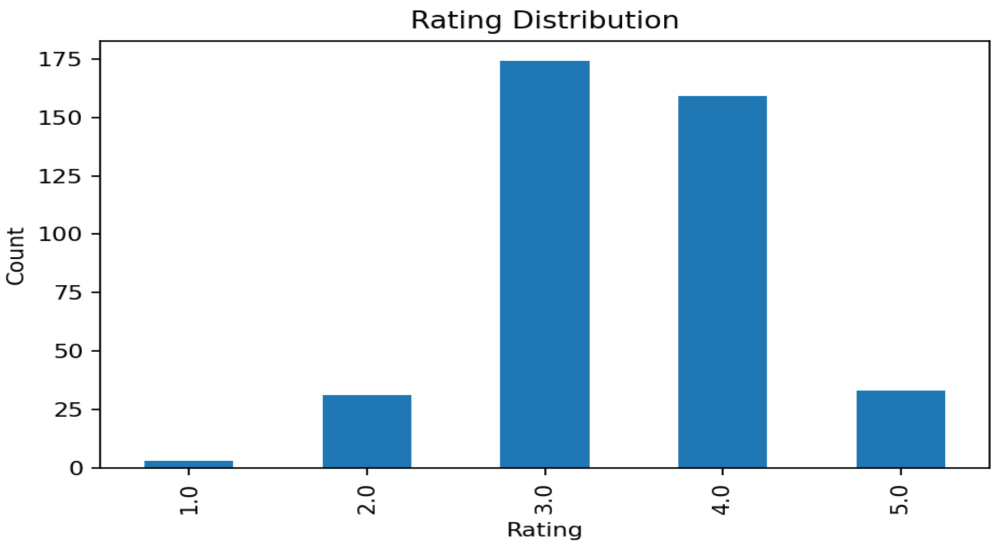
Problem Understanding

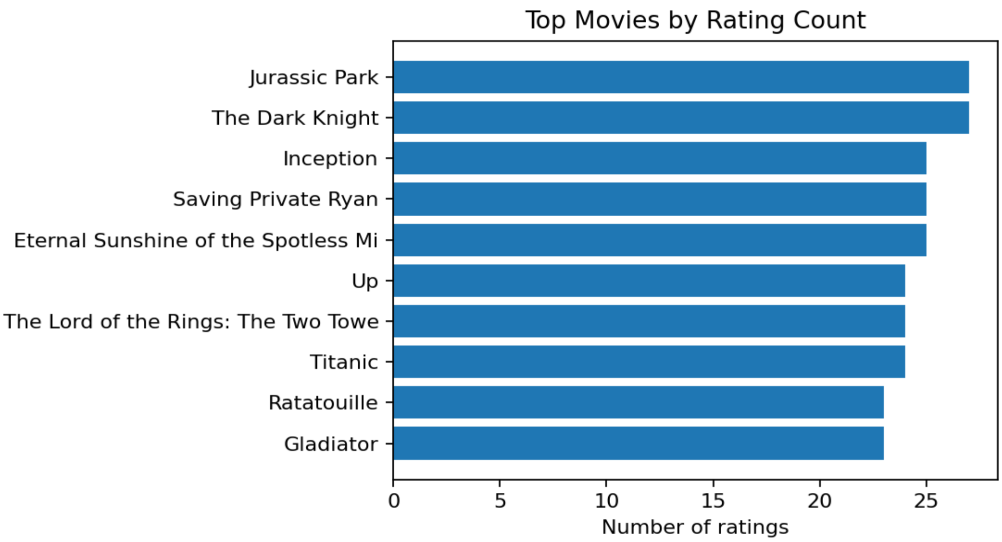
The objective is to learn user preferences from historical user-movie ratings, predict unseen ratings, and generate useful Top-K content recommendations. The solution must balance rating-prediction accuracy with ranking quality, because a streaming platform ultimately needs relevant movies near the top of the recommendation list.

Dataset and EDA

Metric	Value
Ratings	400
Users	40
Movies	18
Mean rating	3.470
Sparsity	44.44%
Date range	2003-01-21 to 2006-04-26

The Netflix-style recommendation problem is highly sparse: users rate only a small subset of movies. This makes naive popularity recommendations insufficient and motivates collaborative filtering and latent-factor approaches.




Top Movies by Rating Count
Jurassic Park
The Dark Knight
Inception
Saving Private Ryan
Eternal Sunshine of the Spotless Mi
Up
The Lord of the Rings: The Two Towe
Titanic
Ratatouille
Gladiator
0
5
10
15
20
25
Number of ratings

Methodology

- Bias baseline: global mean + user bias + movie bias. This is the sanity-check model.
- Item-based collaborative filtering: computes movie-to-movie similarity from rating patterns and recommends movies similar to those the user has rated highly.
- Matrix factorization: learns latent user and movie vectors using stochastic gradient descent with bias terms.
- Temporal train-test split: older ratings are used for training and newer ratings are held out for testing, avoiding random time leakage.

Evaluation Strategy

RMSE and MAE evaluate predicted rating accuracy. MAP@10 evaluates whether relevant held-out movies appear near the top of each user's recommendation list. A movie is treated as relevant when its actual rating is greater than or equal to 3.5.

Experimental Results

model	RMSE	MAE	MAP@10	P@10	R@10	Hit@10	Users
item_item_cf	0.8136	0.6131	0.4188	0.1444	0.9630	1.0000	27
matrix_factorization_sgd	0.6442	0.5201	0.4162	0.1481	0.9815	1.0000	27
bias_baseline	0.6444	0.5534	0.3775	0.1481	0.9815	1.0000	27

On the bundled demo dataset, the strongest ranking model is item_item_cf with MAP@10=0.4188. For the full Netflix dataset, these numbers should be regenerated after increasing the training subset size.

Recommendation Generation and Explainability

user_id	rank	movie_id	title	score
1	1	2	The Shawshank Redemption	3.988
1	2	18	Batman Begins	3.910
1	3	8	Gladiator	3.859
1	4	1	The Matrix	3.810
1	5	15	Ratatouille	3.785
1	6	9	Titanic	3.678
1	7	13	Eternal Sunshine of the Spotless Mind	3.631
1	8	10	The Lord of the Rings: The Two Towers	3.594
2	1	3	Spirited Away	3.852
2	2	18	Batman Begins	3.738

The item-based model supports explanations by identifying previously rated movies that are most similar to a recommended movie. This improves transparency and makes the dashboard more useful for judging recommendation quality.

Key Insights

- Sparse interactions mean that filtering very inactive users/items improves small-scale experiments.
- A bias baseline is hard to beat on RMSE, so ranking metrics are necessary to judge actual recommendation usefulness.
- Item-based collaborative filtering gives strong explainability and is practical for a dashboard prototype.
- Matrix factorization is scalable in concept but needs careful tuning and larger data to show its full advantage.

Failure Cases and Limitations

- Cold-start users and movies cannot be personalized without additional metadata or onboarding signals.
- Popularity bias can dominate recommendations when user history is short.
- The demo metrics are not competition metrics; final numbers must be generated on the official Netflix Prize data or a clearly stated subset.

Future Improvements

- Add ensemble blending of baseline, item-CF, and matrix-factorization scores.
- Add diversity re-ranking so the top list is not too narrow.
- Use movie metadata for cold-start and explainable hybrid recommendations.
- Scale evaluation with sampled candidate sets and batch scoring for larger Netflix subsets.

Conclusion

The project delivers a complete recommender-system pipeline with EDA, model comparison, RMSE, MAP@10, Top-K recommendations, explainability, and a dashboard. The design is simple enough to reproduce but strong enough to defend technically.